

Manual Change History (3666214) QST30 Industrial Electronic Control System Troubleshooting and Repair Manual

Date	Not Available
Section	front
Procedure Number	3666214-toc (/qs3/pubsys2/xml/en/manual/3666214/3666214-toc.html)
Procedure Title	table of contents
Update Reason	
Date	12-Mar-2002
Section	Section i - Introduction
Procedure Number	99-204-004 (/qs3/pubsys2/xml/en/procedures/99/99-204-004.html)
Procedure Title	Symbols
Update Reason	
Date	12-Mar-2002
Section	Section i - Introduction
Procedure Number	99-204-005 (/qs3/pubsys2/xml/en/procedures/99/99-204-005.html)
Procedure Title	Illustrations
Update Reason	
Date	12-Mar-2002
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-361 (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html)
Procedure Title	Component Connector and Pin Inspection
Update Reason	

Date	03-Apr-2002
Section	Section TS - Troubleshooting Symptoms
Procedure Number	99-t00-002 (/qs3/pubsys2/xml/en/procedures/99/99-t00-002.html)
Procedure Title	Troubleshooting Symptom Charts
Update Reason	
Date	10-Jul-2002
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	87-022-001_19 (/qs3/pubsys2/xml/en/procedures/87/87-022-001_19.html)
Procedure Title	Service Tools
Update Reason	
Date	10-Jul-2002
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-220 (/qs3/pubsys2/xml/en/procedures/99/99-019-220.html)
Procedure Title	Augat Connector Series
Update Reason	
Date	22-Jan-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-052 (/qs3/pubsys2/xml/en/procedures/99/99-019-052.html)
Procedure Title	Idle Adjust Switch
Update Reason	

Date	10-Feb-2003
Section	Section F - Familiarization
Procedure Number	87-101-007 (/qs3/pubsys2/xml/en/procedures/87/87-101-007.html)
Procedure Title	Electronic Controlled Fuel System
Update Reason	

Date	10-Feb-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	87-019-011 (/qs3/pubsys2/xml/en/procedures/87/87-019-011.html)
Procedure Title	Blowby Pressure Sensor
Update Reason	

Date	10-Feb-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	87-019-016 (/qs3/pubsys2/xml/en/procedures/87/87-019-016.html)
Procedure Title	Coolant Pressure Sensor
Update Reason	

Date	10-Feb-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	87-019-017 (/qs3/pubsys2/xml/en/procedures/87/87-019-017.html)
Procedure Title	Engine Coolant Level Sensor
Update Reason	

Date	10-Feb-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	87-019-019 (/qs3/pubsys2/xml/en/procedures/87/87-019-019.html)
Procedure Title	Engine Coolant Temperature Sensor
Update Reason	
Date	10-Feb-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	87-019-031 (/qs3/pubsys2/xml/en/procedures/87/87-019-031.html)
Procedure Title	Electronic Control Module (ECM)
Update Reason	
Date	10-Feb-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	87-019-038 (/qs3/pubsys2/xml/en/procedures/87/87-019-038.html)
Procedure Title	Engine Position Sensor (EPS)
Update Reason	
Date	10-Feb-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	87-019-039 (/qs3/pubsys2/xml/en/procedures/87/87-019-039.html)
Procedure Title	Engine Position Sensor (EPS) Circuit
Update Reason	

Date	10-Feb-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	87-019-040 (/qs3/pubsys2/xml/en/procedures/87/87-019-040.html)
Procedure Title	Engine Protection Warning Lamp
Update Reason	
Date	10-Feb-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	87-019-041 (/qs3/pubsys2/xml/en/procedures/87/87-019-041.html)
Procedure Title	Engine Protection Warning Lamp Circuit
Update Reason	
Date	10-Feb-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	87-019-042 (/qs3/pubsys2/xml/en/procedures/87/87-019-042.html)
Procedure Title	Engine Speed Sensor (ESS)
Update Reason	
Date	10-Feb-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	87-019-043 (/qs3/pubsys2/xml/en/procedures/87/87-019-043.html)
Procedure Title	Engine Wiring Harness
Update Reason	

Date	10-Feb-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	87-019-050 (/qs3/pubsys2/xml/en/procedures/87/87-019-050.html)
Procedure Title	Fuel Shutoff Valve
Update Reason	
Date	10-Feb-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	87-019-061 (/qs3/pubsys2/xml/en/procedures/87/87-019-061.html)
Procedure Title	Intake Manifold Pressure Sensor
Update Reason	
Date	10-Feb-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	87-019-066 (/qs3/pubsys2/xml/en/procedures/87/87-019-066.html)
Procedure Title	Engine Oil Pressure Sensor
Update Reason	
Date	10-Feb-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	87-019-067 (/qs3/pubsys2/xml/en/procedures/87/87-019-067.html)
Procedure Title	Engine Oil Temperature Sensor
Update Reason	

Date	10-Feb-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	87-019-086 (/qs3/pubsys2/xml/en/procedures/87/87-019-086.html)
Procedure Title	Accelerator Pedal or Lever Position Sensor Circuit
Update Reason	

Date	10-Feb-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	87-019-087 (/qs3/pubsys2/xml/en/procedures/87/87-019-087.html)
Procedure Title	Unswitched Battery Supply Circuit
Update Reason	

Date	10-Feb-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	87-019-106 (/qs3/pubsys2/xml/en/procedures/87/87-019-106.html)
Procedure Title	Engine Speed Sensor Circuit
Update Reason	

Date	10-Feb-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	87-019-107 (/qs3/pubsys2/xml/en/procedures/87/87-019-107.html)
Procedure Title	Intermediate Speed Control Switch
Update Reason	

Date	10-Feb-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	87-019-108 (/qs3/pubsys2/xml/en/procedures/87/87-019-108.html)
Procedure Title	Intermediate Speed Control Validation Switch
Update Reason	
Date	10-Feb-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	87-019-174 (/qs3/pubsys2/xml/en/procedures/87/87-019-174.html)
Procedure Title	Backbone Wiring Harness
Update Reason	
Date	10-Feb-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	87-019-195 (/qs3/pubsys2/xml/en/procedures/87/87-019-195.html)
Procedure Title	Connector, Fuse Holder
Update Reason	
Date	10-Feb-2003
Section	Section L - Service Literature
Procedure Number	87-205-001 (/qs3/pubsys2/xml/en/procedures/87/87-205-001.html)
Procedure Title	Additional Service Literature
Update Reason	

Date	05-Mar-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	87-022-015 (/qs3/pubsys2/xml/en/procedures/87/87-022-015.html)
Procedure Title	Wiring Harness Repair Kit
Update Reason	
Date	18-Mar-2003
Section	front
Procedure Number	3666214-titlepage (/qs3/pubsys2/xml/en/manual/3666214/3666214-titlepage.html)
Procedure Title	titlepage
Update Reason	
Date	03-Apr-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-027 (/qs3/pubsys2/xml/en/procedures/99/99-019-027.html)
Procedure Title	Diagnostic Test Mode Switch
Update Reason	
Date	03-Apr-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-046 (/qs3/pubsys2/xml/en/procedures/99/99-019-046.html)
Procedure Title	Fault Lamp
Update Reason	

Date	03-Apr-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-085 (/qs3/pubsys2/xml/en/procedures/99/99-019-085.html)
Procedure Title	Accelerator Pedal or Lever Position Sensor
Update Reason	
Date	03-Apr-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-197 (/qs3/pubsys2/xml/en/procedures/99/99-019-197.html)
Procedure Title	Ring Terminal
Update Reason	
Date	03-Apr-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-198 (/qs3/pubsys2/xml/en/procedures/99/99-019-198.html)
Procedure Title	Fuse, Harness In-Line
Update Reason	
Date	03-Apr-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-199 (/qs3/pubsys2/xml/en/procedures/99/99-019-199.html)
Procedure Title	Connector, Butt-Splice
Update Reason	

Date	03-Apr-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-202 (/qs3/pubsys2/xml/en/procedures/99/99-019-202.html)
Procedure Title	Metripack Connector Series
Update Reason	
Date	03-Apr-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-203 (/qs3/pubsys2/xml/en/procedures/99/99-019-203.html)
Procedure Title	AMP Connector Series
Update Reason	
Date	03-Apr-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-204 (/qs3/pubsys2/xml/en/procedures/99/99-019-204.html)
Procedure Title	Deutsch DRC Connector Series
Update Reason	
Date	03-Apr-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-206 (/qs3/pubsys2/xml/en/procedures/99/99-019-206.html)
Procedure Title	Deutsch DTM and DTP Connector Series
Update Reason	

Date	03-Apr-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-209 (/qs3/pubsys2/xml/en/procedures/99/99-019-209.html)
Procedure Title	ITT Cannon Connector Series
Update Reason	
Date	28-May-2003
Section	Section E - System Identification
Procedure Number	87-100-001 (/qs3/pubsys2/xml/en/procedures/87/87-100-001.html)
Procedure Title	Engine Identification
Update Reason	
Date	09-Oct-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-201 (/qs3/pubsys2/xml/en/procedures/99/99-019-201.html)
Procedure Title	Weather Pak Connector Series
Update Reason	
Date	09-Oct-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-205 (/qs3/pubsys2/xml/en/procedures/99/99-019-205.html)
Procedure Title	Deutsch DT Connector Series
Update Reason	

Date	09-Oct-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-207 (/qs3/pubsys2/xml/en/procedures/99/99-019-207.html)
Procedure Title	Deutsch HD10 Connector Series
Update Reason	
Date	09-Oct-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-210 (/qs3/pubsys2/xml/en/procedures/99/99-019-210.html)
Procedure Title	AMP VP44 Connector Series
Update Reason	
Date	09-Oct-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-211 (/qs3/pubsys2/xml/en/procedures/99/99-019-211.html)
Procedure Title	Packard Relay Connector
Update Reason	
Date	09-Oct-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-212 (/qs3/pubsys2/xml/en/procedures/99/99-019-212.html)
Procedure Title	Ford™ Connector Series
Update Reason	

Date	09-Oct-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-213 (/qs3/pubsys2/xml/en/procedures/99/99-019-213.html)
Procedure Title	D-Sub Miniature Connector
Update Reason	
Date	09-Oct-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-218 (/qs3/pubsys2/xml/en/procedures/99/99-019-218.html)
Procedure Title	Framatome Connector Series
Update Reason	
Date	21-Oct-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc113 (/qs3/pubsys2/xml/en/procedures/87/87-fc113.html)
Procedure Title	Timing Sleeve Actuator Circuit
Update Reason	
Date	21-Oct-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc114 (/qs3/pubsys2/xml/en/procedures/87/87-fc114.html)
Procedure Title	Timing Sleeve Actuator Circuit
Update Reason	
Date	21-Oct-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc121 (/qs3/pubsys2/xml/en/procedures/87/87-fc121.html)
Procedure Title	Engine Speed Sensor Circuit
Update Reason	

Date	21-Oct-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc122 (/qs3/pubsys2/xml/en/procedures/87/87-fc122.html)
Procedure Title	Intake Manifold Pressure Sensor Circuit
Update Reason	

Date	21-Oct-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc123 (/qs3/pubsys2/xml/en/procedures/87/87-fc123.html)
Procedure Title	Intake Manifold Pressure Sensor Circuit
Update Reason	

Date	21-Oct-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc131 (/qs3/pubsys2/xml/en/procedures/87/87-fc131.html)
Procedure Title	Accelerator Pedal or Lever Position Sensor Circuit
Update Reason	

Date	21-Oct-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc132 (/qs3/pubsys2/xml/en/procedures/87/87-fc132.html)
Procedure Title	Accelerator Pedal or Lever Position Sensor Circuit
Update Reason	

Date	21-Oct-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc133 (/qs3/pubsys2/xml/en/procedures/87/87-fc133.html)
Procedure Title	Remote Accelerator Pedal or Lever Position Sensor Circuit
Update Reason	
Date	21-Oct-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc134 (/qs3/pubsys2/xml/en/procedures/87/87-fc134.html)
Procedure Title	Remote Accelerator Pedal or Lever Position Sensor Circuit
Update Reason	
Date	21-Oct-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc135 (/qs3/pubsys2/xml/en/procedures/87/87-fc135.html)
Procedure Title	Oil Pressure Sensor Circuit
Update Reason	
Date	21-Oct-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc141 (/qs3/pubsys2/xml/en/procedures/87/87-fc141.html)
Procedure Title	Oil Pressure Sensor Circuit
Update Reason	
Date	21-Oct-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc144 (/qs3/pubsys2/xml/en/procedures/87/87-fc144.html)
Procedure Title	Coolant Temperature Sensor Circuit
Update Reason	

Date	21-Oct-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc145 (/qs3/pubsys2/xml/en/procedures/87/87-fc145.html)
Procedure Title	Coolant Temperature Sensor Circuit
Update Reason	
Date	21-Oct-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc148 (/qs3/pubsys2/xml/en/procedures/87/87-fc148.html)
Procedure Title	Frequency Accelerator Pedal or Lever Position Signal Circuit
Update Reason	
Date	21-Oct-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc154 (/qs3/pubsys2/xml/en/procedures/87/87-fc154.html)
Procedure Title	Intake Manifold Temperature Sensor Circuit
Update Reason	
Date	21-Oct-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc184 (/qs3/pubsys2/xml/en/procedures/87/87-fc184.html)
Procedure Title	Primary or Secondary Electronic Control Module (ECM) Identification Circuit Error
Update Reason	

Date	21-Oct-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc185 (/qs3/pubsys2/xml/en/procedures/87/87-fc185.html)
Procedure Title	Primary Electronic Control Module (ECM) to Secondary ECM Communication Error
Update Reason	
Date	21-Oct-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc212 (/qs3/pubsys2/xml/en/procedures/87/87-fc212.html)
Procedure Title	Oil Temperature Sensor Circuit
Update Reason	
Date	21-Oct-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc213 (/qs3/pubsys2/xml/en/procedures/87/87-fc213.html)
Procedure Title	Oil Temperature Sensor Circuit
Update Reason	
Date	21-Oct-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc221 (/qs3/pubsys2/xml/en/procedures/87/87-fc221.html)
Procedure Title	Barometric Air Pressure Sensor Circuit
Update Reason	
Date	21-Oct-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc222 (/qs3/pubsys2/xml/en/procedures/87/87-fc222.html)
Procedure Title	Barometric Air Pressure Sensor Circuit
Update Reason	

Date	21-Oct-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc232 (/qs3/pubsys2/xml/en/procedures/87/87-fc232.html)
Procedure Title	Coolant Pressure Sensor Circuit
Update Reason	
Date	21-Oct-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc254 (/qs3/pubsys2/xml/en/procedures/87/87-fc254.html)
Procedure Title	Fuel Shutoff Valve Circuit
Update Reason	
Date	21-Oct-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc259 (/qs3/pubsys2/xml/en/procedures/87/87-fc259.html)
Procedure Title	Fuel Shutoff Valve Circuit
Update Reason	
Date	21-Oct-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc346 (/qs3/pubsys2/xml/en/procedures/87/87-fc346.html)
Procedure Title	Electronic Control Module (ECM) Power Down Error
Update Reason	
Date	21-Oct-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc422 (/qs3/pubsys2/xml/en/procedures/87/87-fc422.html)
Procedure Title	Coolant Level Sensor Circuit
Update Reason	

Date	23-Oct-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc111 (/qs3/pubsys2/xml/en/procedures/87/87-fc111.html)
Procedure Title	Electronic Control Module (ECM) Microprocessor
Update Reason	
Date	23-Oct-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc143 (/qs3/pubsys2/xml/en/procedures/87/87-fc143.html)
Procedure Title	Oil Pressure - Engine Protection
Update Reason	
Date	23-Oct-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc151 (/qs3/pubsys2/xml/en/procedures/87/87-fc151.html)
Procedure Title	Coolant Temperature - Engine Protection
Update Reason	
Date	23-Oct-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc155 (/qs3/pubsys2/xml/en/procedures/87/87-fc155.html)
Procedure Title	Intake Manifold Pressure/Temperature - Engine Protection
Update Reason	
Date	23-Oct-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc172 (/qs3/pubsys2/xml/en/procedures/87/87-fc172.html)
Procedure Title	Fuel Control Rack Sticking
Update Reason	

Date	23-Oct-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc173 (/qs3/pubsys2/xml/en/procedures/87/87-fc173.html)
Procedure Title	Fuel Control Rack Sticking
Update Reason	

Date	23-Oct-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc233 (/qs3/pubsys2/xml/en/procedures/87/87-fc233.html)
Procedure Title	Coolant Pressure - Engine Protection
Update Reason	

Date	23-Oct-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc234 (/qs3/pubsys2/xml/en/procedures/87/87-fc234.html)
Procedure Title	Engine Overspeed
Update Reason	

Date	23-Oct-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc235 (/qs3/pubsys2/xml/en/procedures/87/87-fc235.html)
Procedure Title	Engine Coolant Level - Engine Protection
Update Reason	

Date	23-Oct-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc719 (/qs3/pubsys2/xml/en/procedures/87/87-fc719.html)
Procedure Title	Crankcase Pressure Sensor Circuit
Update Reason	

Date	23-Oct-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc729 (/qs3/pubsys2/xml/en/procedures/87/87-fc729.html)
Procedure Title	Crankcase Pressure Sensor Circuit
Update Reason	
Date	23-Oct-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fcmulti (/qs3/pubsys2/xml/en/procedures/87/87-fcmulti.html)
Procedure Title	Multiple Fault Codes
Update Reason	
Date	27-Oct-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc115 (/qs3/pubsys2/xml/en/procedures/87/87-fc115.html)
Procedure Title	Engine Speed Sensor Circuit
Update Reason	
Date	27-Oct-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc147 (/qs3/pubsys2/xml/en/procedures/87/87-fc147.html)
Procedure Title	Frequency Accelerator Pedal or Lever Position Signal Circuit
Update Reason	
Date	27-Oct-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc153 (/qs3/pubsys2/xml/en/procedures/87/87-fc153.html)
Procedure Title	Intake Manifold Temperature Sensor Circuit
Update Reason	

Date	27-Oct-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc231 (/qs3/pubsys2/xml/en/procedures/87/87-fc231.html)
Procedure Title	Coolant Pressure Sensor Circuit
Update Reason	
Date	27-Oct-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc342 (/qs3/pubsys2/xml/en/procedures/87/87-fc342.html)
Procedure Title	Electronic Control Module (ECM) Calibration Error
Update Reason	
Date	27-Oct-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc343 (/qs3/pubsys2/xml/en/procedures/87/87-fc343.html)
Procedure Title	Electronic Control Module (ECM) Internal Communication Error
Update Reason	
Date	27-Oct-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc415 (/qs3/pubsys2/xml/en/procedures/87/87-fc415.html)
Procedure Title	Oil Pressure - Engine Protection
Update Reason	
Date	27-Oct-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc423 (/qs3/pubsys2/xml/en/procedures/87/87-fc423.html)
Procedure Title	Timing Sleeve Actuator Sticking
Update Reason	

Date	27-Oct-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc431 (/qs3/pubsys2/xml/en/procedures/87/87-fc431.html)
Procedure Title	Idle Validation Switch Circuit
Update Reason	
Date	27-Oct-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc432 (/qs3/pubsys2/xml/en/procedures/87/87-fc432.html)
Procedure Title	Accelerator Pedal or Lever Position Sensor and Idle Validation Switch Circuit
Update Reason	
Date	27-Oct-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc441 (/qs3/pubsys2/xml/en/procedures/87/87-fc441.html)
Procedure Title	Unswitched Battery Supply Circuit
Update Reason	
Date	27-Oct-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc442 (/qs3/pubsys2/xml/en/procedures/87/87-fc442.html)
Procedure Title	Unswitched Battery Supply Circuit
Update Reason	
Date	27-Oct-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc524 (/qs3/pubsys2/xml/en/procedures/87/87-fc524.html)
Procedure Title	Alternate Droop Switch Circuit
Update Reason	

Date	27-Oct-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc527 (/qs3/pubsys2/xml/en/procedures/87/87-fc527.html)
Procedure Title	Solenoid A Circuit Error
Update Reason	
Date	27-Oct-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc528 (/qs3/pubsys2/xml/en/procedures/87/87-fc528.html)
Procedure Title	Alternate Torque Switch Circuit
Update Reason	
Date	27-Oct-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc529 (/qs3/pubsys2/xml/en/procedures/87/87-fc529.html)
Procedure Title	Solenoid B Circuit Error
Update Reason	
Date	27-Oct-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc551 (/qs3/pubsys2/xml/en/procedures/87/87-fc551.html)
Procedure Title	Idle Validation Switch Circuit
Update Reason	
Date	27-Oct-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc555 (/qs3/pubsys2/xml/en/procedures/87/87-fc555.html)
Procedure Title	Crankcase Pressure - Engine Protection
Update Reason	

Date	05-Feb-2004
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666214-t066-tr (/qs3/pubsys2/xml/en/manual/3666214/3666214-t066-tr.html)
Procedure Title	Engine Speed Surges at Low or High Idle
Update Reason	
Date	05-Feb-2004
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666214-t067-tr (/qs3/pubsys2/xml/en/manual/3666214/3666214-t067-tr.html)
Procedure Title	Engine Speed Surges Under Load or in Operating Range
Update Reason	
Date	05-Feb-2004
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666214-t072-tr (/qs3/pubsys2/xml/en/manual/3666214/3666214-t072-tr.html)
Procedure Title	Engine Starts But Will Not Keep Running
Update Reason	
Date	05-Feb-2004
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666214-t080-tr (/qs3/pubsys2/xml/en/manual/3666214/3666214-t080-tr.html)
Procedure Title	Engine Will Not Reach Rated Speed (RPM)
Update Reason	
Date	05-Feb-2004
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666214-t081-tr (/qs3/pubsys2/xml/en/manual/3666214/3666214-t081-tr.html)
Procedure Title	Engine Will Not Shut Off
Update Reason	

Date	05-Feb-2004
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666214-t083-tr (/qs3/pubsys2/xml/en/manual/3666214/3666214-t083-tr.html)
Procedure Title	Fault Code Warning Lamps Stay On (No Apparent Reason)
Update Reason	
Date	05-Feb-2004
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666214-t084-tr (/qs3/pubsys2/xml/en/manual/3666214/3666214-t084-tr.html)
Procedure Title	Fault Code Warning Lamps Do Not Illuminate
Update Reason	
Date	05-Feb-2004
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666214-t087-tr (/qs3/pubsys2/xml/en/manual/3666214/3666214-t087-tr.html)
Procedure Title	Fuel Consumption Excessive
Update Reason	
Date	05-Feb-2004
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666214-t099-tr (/qs3/pubsys2/xml/en/manual/3666214/3666214-t099-tr.html)
Procedure Title	Low Idle Adjust Switch Does Not Work
Update Reason	
Date	05-Feb-2004
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666214-t116-tr (/qs3/pubsys2/xml/en/manual/3666214/3666214-t116-tr.html)
Procedure Title	Smoke, Black-Excessive
Update Reason	

Date	05-Feb-2004
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666214-t118-tr (/qs3/pubsys2/xml/en/manual/3666214/3666214-t118-tr.html)
Procedure Title	Smoke, White-Excessive
Update Reason	
Date	06-Feb-2004
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666214-t033-tr (/qs3/pubsys2/xml/en/manual/3666214/3666214-t033-tr.html)
Procedure Title	Engine Acceleration or Response Poor
Update Reason	
Date	06-Feb-2004
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666214-t041-tr (/qs3/pubsys2/xml/en/manual/3666214/3666214-t041-tr.html)
Procedure Title	Engine Decelerates Slowly
Update Reason	
Date	06-Feb-2004
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666214-t043-tr (/qs3/pubsys2/xml/en/manual/3666214/3666214-t043-tr.html)
Procedure Title	Engine Difficult to Start or Will Not Start (Exhaust Smoke)
Update Reason	
Date	06-Feb-2004
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666214-t059-tr (/qs3/pubsys2/xml/en/manual/3666214/3666214-t059-tr.html)
Procedure Title	Engine Runs on One Bank Only
Update Reason	

Date	06-Feb-2004
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666214-t061-tr (/qs3/pubsys2/xml/en/manual/3666214/3666214-t061-tr.html)
Procedure Title	Engine Runs Rough at Idle
Update Reason	
Date	06-Feb-2004
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666214-t062-tr (/qs3/pubsys2/xml/en/manual/3666214/3666214-t062-tr.html)
Procedure Title	Engine Runs Rough or Misfires
Update Reason	
Date	06-Feb-2004
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666214-t064-tr (/qs3/pubsys2/xml/en/manual/3666214/3666214-t064-tr.html)
Procedure Title	Engine Shuts Off Unexpectedly or Dies During Deceleration
Update Reason	
Date	06-Feb-2004
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	87-019-059 (/qs3/pubsys2/xml/en/procedures/87/87-019-059.html)
Procedure Title	Intake Manifold Air Temperature Sensor
Update Reason	
Date	03-Mar-2004
Section	Section V - Specifications
Procedure Number	99-018-010 (/qs3/pubsys2/xml/en/procedures/99/99-018-010.html)
Procedure Title	Fraction, Decimal, Millimeter Conversions
Update Reason	

Date	03-Mar-2004
Section	Section V - Specifications
Procedure Number	99-018-011 (/qs3/pubsys2/xml/en/procedures/99/99-018-011.html)
Procedure Title	Newton-Meter to Foot-Pound Conversions
Update Reason	
Date	03-Mar-2004
Section	Section V - Specifications
Procedure Number	99-018-012 (/qs3/pubsys2/xml/en/procedures/99/99-018-012.html)
Procedure Title	Pipe Plug Torque Values
Update Reason	
Date	03-Mar-2004
Section	Section V - Specifications
Procedure Number	99-018-013 (/qs3/pubsys2/xml/en/procedures/99/99-018-013.html)
Procedure Title	Tap-Drill Chart - U.S. Customary and Metric
Update Reason	
Date	03-Mar-2004
Section	Section V - Specifications
Procedure Number	99-018-014 (/qs3/pubsys2/xml/en/procedures/99/99-018-014.html)
Procedure Title	Weights and Measures - Conversion Factors
Update Reason	
Date	09-Jun-2004
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-208 (/qs3/pubsys2/xml/en/procedures/99/99-019-208.html)
Procedure Title	Deutsch HDP20 and HD30 Connector Series
Update Reason	

Date	09-Aug-2004
Section	Section E - System Identification
Procedure Number	87-100-002 (/qs3/pubsys2/xml/en/procedures/87/87-100-002.html)
Procedure Title	Engine Diagrams
Update Reason	
Date	09-Aug-2004
Section	back
Procedure Number	3666214-backcover (/qs3/pubsys2/xml/en/manual/3666214/3666214-backcover.html)
Procedure Title	Backcover
Update Reason	
Date	24-Aug-2004
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	87-019-004 (/qs3/pubsys2/xml/en/procedures/87/87-019-004.html)
Procedure Title	Barometric Air Pressure Sensor
Update Reason	
Date	06-Feb-2006
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	105-019-428 (/qs3/pubsys2/xml/en/procedures/105/105-019-428.html)
Procedure Title	Engine Datalinks
Update Reason	
Date	22-May-2006
Section	front
Procedure Number	tr-frwd (/qs3/pubsys2/xml/en/manual/preface/tr-frwd.html)
Procedure Title	foreword
Update Reason	

Date	27-Jun-2006
Section	Section L - Service Literature
Procedure Number	99-205-002 (/qs3/pubsys2/xml/en/procedures/99/99-205-002.html)
Procedure Title	Cummins Customized Parts Catalog
Update Reason	
Date	16-Jan-2007
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc773 (/qs3/pubsys2/xml/en/procedures/87/87-fc773.html)
Procedure Title	Right Bank Needle Lift Sensor Reading Is Not Being Detected or Is Out of Range
Update Reason	
Date	08-May-2007
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc2265 (/qs3/pubsys2/xml/en/procedures/87/87-fc2265.html)
Procedure Title	Electric Lift Pump for Engine Fuel Supply Circuit - Voltage Above Normal or Shorted to High Source
Update Reason	
Date	08-May-2007
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc2266 (/qs3/pubsys2/xml/en/procedures/87/87-fc2266.html)
Procedure Title	Electric Lift Pump for Engine Fuel Supply Circuit - Voltage Below Normal or Shorted to Low Source
Update Reason	

Date	21-May-2007
Section	Section V - Specifications
Procedure Number	4021638 (/qs3/pubsys2/xml/en/procedures//4021638.html)
Procedure Title	QST30 One-Piece Harness Wiring Diagram Specifications
Update Reason	
Date	24-Jul-2007
Section	Section i - Introduction
Procedure Number	99-204-008 (/qs3/pubsys2/xml/en/procedures/99/99-204-008.html)
Procedure Title	General Cleaning Instructions
Update Reason	
Date	22-Oct-2007
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc166 (/qs3/pubsys2/xml/en/procedures/87/87-fc166.html)
Procedure Title	Rack Actuator Circuit
Update Reason	
Date	07-Dec-2007
Section	Section TT - Troubleshooting Symptoms (New Format)
Procedure Number	00-ttecm (/qs3/pubsys2/xml/en/procedures/00/00-ttecm.html)
Procedure Title	ECM - No Communication Troubleshooting Tree
Update Reason	

Date	28-Jan-2008
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-360 (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html)
Procedure Title	Resistance Measurement Using a Multimeter and a Wiring Diagram
Update Reason	

Date	13-Feb-2008
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666214-t057-tr (/qs3/pubsys2/xml/en/manual/3666214/3666214-t057-tr.html)
Procedure Title	Engine Power Output Low
Update Reason	

Date	30-May-2008
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-008 (/qs3/pubsys2/xml/en/procedures/99/99-019-008.html)
Procedure Title	Battery Ground Circuit
Update Reason	

Date	30-May-2008
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-028 (/qs3/pubsys2/xml/en/procedures/99/99-019-028.html)
Procedure Title	Diagnostic Test Mode Switch Circuit
Update Reason	

Date	30-May-2008
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-047 (/qs3/pubsys2/xml/en/procedures/99/99-019-047.html)
Procedure Title	Fault Lamp Circuit
Update Reason	
Date	30-May-2008
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-053 (/qs3/pubsys2/xml/en/procedures/99/99-019-053.html)
Procedure Title	Idle Adjust Switch Circuit
Update Reason	
Date	30-May-2008
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-071 (/qs3/pubsys2/xml/en/procedures/99/99-019-071.html)
Procedure Title	OEM Wiring Harness
Update Reason	
Date	19-Mar-2009
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-215 (/qs3/pubsys2/xml/en/procedures/99/99-019-215.html)
Procedure Title	Bosch® ECM Injector Driver Connector Series
Update Reason	

Date	08-Apr-2009
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc583 (/qs3/pubsys2/xml/en/procedures/87/87-fc583.html)
Procedure Title	Fuel Inlet Restriction Sensor Circuit
Update Reason	
Date	23-Apr-2009
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-216 (/qs3/pubsys2/xml/en/procedures/99/99-019-216.html)
Procedure Title	Bosch® ECM Actuator and Sensor Connector Series
Update Reason	
Date	23-Apr-2009
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-217 (/qs3/pubsys2/xml/en/procedures/99/99-019-217.html)
Procedure Title	Bosch® ECM OEM Connector Series
Update Reason	
Date	25-Jun-2009
Section	Section TS - Troubleshooting Symptoms
Procedure Number	99-t00-001-tr (/qs3/pubsys2/xml/en/procedures/99/99-t00-001-tr.html)
Procedure Title	Troubleshooting Procedures and Techniques
Update Reason	

Date	22-Jul-2009
Section	Section L - Service Literature
Procedure Number	99-205-004 (/qs3/pubsys2/xml/en/procedures/99/99-205-004.html)
Procedure Title	Service Literature Ordering Location
Update Reason	
Date	21-Aug-2009
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc772 (/qs3/pubsys2/xml/en/procedures/87/87-fc772.html)
Procedure Title	Left Bank Needle Lift Sensor Reading Is Not Being Detected or Is Out of Range
Update Reason	
Date	02-Sep-2009
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc214 (/qs3/pubsys2/xml/en/procedures/87/87-fc214.html)
Procedure Title	Engine Oil Temperature High – Critical
Update Reason	
Date	02-Sep-2009
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc219 (/qs3/pubsys2/xml/en/procedures/87/87-fc219.html)
Procedure Title	Low Oil Level Detected in Centinel™ Make Up Tank
Update Reason	

Date	02-Sep-2009
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc223 (/qs3/pubsys2/xml/en/procedures/87/87-fc223.html)
Procedure Title	Centinel® Burn Valve Solenoid Circuit - Voltage Incorrect
Update Reason	
Date	02-Sep-2009
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc261 (/qs3/pubsys2/xml/en/procedures/87/87-fc261.html)
Procedure Title	Fuel Temperature High - Warning
Update Reason	
Date	02-Sep-2009
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc263 (/qs3/pubsys2/xml/en/procedures/87/87-fc263.html)
Procedure Title	Fuel Temperature Sensor Circuit - Shorted High
Update Reason	
Date	02-Sep-2009
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc265 (/qs3/pubsys2/xml/en/procedures/87/87-fc265.html)
Procedure Title	Fuel Temperature Sensor Circuit - Shorted Low
Update Reason	

Date	02-Sep-2009
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc581 (/qs3/pubsys2/xml/en/procedures/87/87-fc581.html)
Procedure Title	Fuel Supply Pressure Sensor Circuit ‐ Shorted High
Update Reason	
Date	02-Sep-2009
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc582 (/qs3/pubsys2/xml/en/procedures/87/87-fc582.html)
Procedure Title	Fuel Supply Pressure Sensor Circuit ‐ Shorted Low
Update Reason	
Date	02-Sep-2009
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc649 (/qs3/pubsys2/xml/en/procedures/87/87-fc649.html)
Procedure Title	Change Lubricating Oil and Filter
Update Reason	
Date	10-Sep-2009
Section	Section i - Introduction
Procedure Number	99-204-002-tr (/qs3/pubsys2/xml/en/procedures/99/99-204-002-tr.html)
Procedure Title	About the Manual
Update Reason	

Date	16-Sep-2009
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-064 (/qs3/pubsys2/xml/en/procedures/99/99-019-064.html)
Procedure Title	Key Switch Battery Supply Circuit
Update Reason	
Date	08-Oct-2009
Section	Section i - Introduction
Procedure Number	99-204-007-tr (/qs3/pubsys2/xml/en/procedures/99/99-204-007-tr.html)
Procedure Title	General Repair Instructions
Update Reason	
Date	02-Dec-2009
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-359 (/qs3/pubsys2/xml/en/procedures/99/99-019-359.html)
Procedure Title	Multimeter Usage
Update Reason	
Date	02-Dec-2009
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-362 (/qs3/pubsys2/xml/en/procedures/99/99-019-362.html)
Procedure Title	Inactive or Intermittent Fault Code
Update Reason	

Date	11-Dec-2009
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666214-t044-tr (/qs3/pubsys2/xml/en/manual/3666214/3666214-t044-tr.html)
Procedure Title	Engine Difficult to Start or Will Not Start (No Exhaust Smoke)
Update Reason	
Date	25-Jan-2010
Section	Section i - Introduction
Procedure Number	99-204-006-tr (/qs3/pubsys2/xml/en/procedures/99/99-204-006-tr.html)
Procedure Title	General Safety Instructions
Update Reason	
Date	25-Jan-2010
Section	Section V - Specifications
Procedure Number	99-018-005 (/qs3/pubsys2/xml/en/procedures/99/99-018-005.html)
Procedure Title	Drive Belt Tension
Update Reason	
Date	25-Jan-2010
Section	Section V - Specifications
Procedure Number	99-018-009 (/qs3/pubsys2/xml/en/procedures/99/99-018-009.html)
Procedure Title	Capscrew Markings and Torque Values
Update Reason	
Date	18-Feb-2010
Section	Section F - Familiarization
Procedure Number	00-022-999 (/qs3/pubsys2/xml/en/procedures/00/00-022-999.html)
Procedure Title	Service Tools and Hardware - Overview
Update Reason	

Date	24-Mar-2010
Section	Section i - Introduction
Procedure Number	99-204-009 (/qs3/pubsys2/xml/en/procedures/99/99-204-009.html)
Procedure Title	Acronyms and Abbreviations
Update Reason	
Date	05-Apr-2010
Section	Section V - Specifications
Procedure Number	3666234 (/qs3/pubsys2/xml/en/procedures//3666234.html)
Procedure Title	Specifications
Update Reason	
Date	25-May-2010
Section	Section i - Introduction
Procedure Number	99-204-003-tr (/qs3/pubsys2/xml/en/procedures/99/99-204-003-tr.html)
Procedure Title	How to Use the Manual
Update Reason	
Date	16-Jun-2010
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	105-019-032 (/qs3/pubsys2/xml/en/procedures/105/105-019-032.html)
Procedure Title	ECM Calibration Code
Update Reason	
Date	23-Jun-2010
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666214-t016-tr (/qs3/pubsys2/xml/en/manual/3666214/3666214-t016-tr.html)
Procedure Title	Communication Error - Electronic Service Tool or Control Device
Update Reason	

Date	20-Oct-2010
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	87-019-043 (/qs3/pubsys2/xml/en/procedures/87/87-019-043.html)
Procedure Title	Engine Wiring Harness
Update Reason	Added the Resistance Check section.
Date	21-Jan-2011
Section	Section F - Familiarization
Procedure Number	00-022-999 (/qs3/pubsys2/xml/en/procedures/00/00-022-999.html)
Procedure Title	Service Tools and Hardware - Overview
Update Reason	Updated General Information, Initial Check, Setup, and Resistance Check sections.
Date	17-Feb-2011
Section	Section 19 - Electronic Engine Controls
Procedure Number	87-019-032 (/qs3/pubsys2/xml/en/procedures/87/87-019-032.html)
Procedure Title	ECM Calibration Code
Update Reason	Add 87-019-032
Date	23-Mar-2011
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc342 (/qs3/pubsys2/xml/en/procedures/87/87-fc342.html)
Procedure Title	Calibration Match Error
Update Reason	Revised Shop Talk section.

Date	28-Jun-2011
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc583 (/qs3/pubsys2/xml/en/procedures/87/87-fc583.html)
Procedure Title	Fuel Pressure Sensor Circuit
Update Reason	Updated fault code for inspection of lift pumps, fuel pressure sensor, and engine wiring harness.

Date	28-Jun-2011
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-t05-583 (/qs3/pubsys2/xml/en/procedures/87/87-t05-583.html)
Procedure Title	Fuel Pressure Sensor Circuit
Update Reason	Updated fault code for inspection of lift pumps, fuel pressure sensor, and engine wiring harness.

Date	30-Apr-2012
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc772 (/qs3/pubsys2/xml/en/procedures/87/87-fc772.html)
Procedure Title	Left Bank Needle Lift Sensor Reading Is Not Being Detected or Is Out of Range
Update Reason	the Lamp was changed from Amber to None and a note was added for information about the earlier calibrations having an amber lamp.

Date	30-Apr-2012
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-t05-772 (/qs3/pubsys2/xml/en/procedures/87/87-t05-772.html)
Procedure Title	Left Bank Needle Lift Sensor Reading Is Not Being Detected or Is Out of Range
Update Reason	Update the fault code troubleshooting tree to the new format with the YES/NO questions. Procedure to replace the number 1 injectors were added.

Date	30-Apr-2012
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc773 (/qs3/pubsys2/xml/en/procedures/87/87-fc773.html)
Procedure Title	Right Bank Needle Lift Sensor Reading Is Not Being Detected or Is Out of Range
Update Reason	the Lamp was changed from Amber to None and a note was added for information about the earlier calibrations having an amber lamp.
Date	30-Apr-2012
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-t05-773 (/qs3/pubsys2/xml/en/procedures/87/87-t05-773.html)
Procedure Title	Right Bank Needle Lift Sensor Reading Is Not Being Detected or Is Out of Range
Update Reason	Update the fault code troubleshooting tree to the new format with the YES/NO questions. Procedure to replace the number 1 injectors were added.
Date	28-Jun-2012
Section	Section 19 - Electronic Engine Controls
Procedure Number	87-022-001_19 (/qs3/pubsys2/xml/en/procedures/87/87-022-001_19.html)
Procedure Title	Service Tools
Update Reason	Added Service Tools: 3163064 (Bench Top Harness), 3885817 (In-Cab Data Link Cable), and 3825141 (In-Cab Data Link Cable).
Date	20-Sep-2013
Section	Section 19 - Electronic Controls - Group 19
Procedure Number	87-019-107 (/qs3/pubsys2/xml/en/procedures/87/87-019-107.html)
Procedure Title	Intermediate Speed Control Switch
Update Reason	Replaced this procedure with a common procedure.

Date	20-Sep-2013
Section	Section 19 - Electronic Controls - Group 19
Procedure Number	00-019-107 (/qs3/pubsys2/xml/en/procedures/00/00-019-107.html)
Procedure Title	Intermediate Speed Control Switch
Update Reason	Added to outline.
Date	20-Jan-2014
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc423 (/qs3/pubsys2/xml/en/procedures/87/87-fc423.html)
Procedure Title	Timing Sleeve Actuator Sticking
Update Reason	Updated fault code tree to include Step 3., Calibration Check.
Date	20-Jan-2014
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-t05-772 (/qs3/pubsys2/xml/en/procedures/87/87-t05-772.html)
Procedure Title	Left Bank Needle Lift Sensor Reading Is Not Being Detected or Is Out of Range
Update Reason	Updated fault code tree to include Step 4., Calibration Check.
Date	20-Jan-2014
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-t05-773 (/qs3/pubsys2/xml/en/procedures/87/87-t05-773.html)
Procedure Title	Right Bank Needle Lift Sensor Reading Is Not Being Detected or Is Out of Range
Update Reason	Updated fault code tree to include Step 4., Calibration Check

Date	08-Jul-2014
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-t05-254 (/qs3/pubsys2/xml/en/procedures/87/87-t05-254.html)
Procedure Title	EHAB (Fuel Shutoff Valve) Circuit
Update Reason	Added Guided Step 1E-4.
Date	05-Dec-2016
Section	Section TT - Troubleshooting Symptoms (New Format)
Procedure Number	00-ttecm (/qs3/pubsys2/xml/en/procedures/00/00-ttecm.html)
Procedure Title	ECM - No Communication Troubleshooting Tree
Update Reason	Modified Steps 2, 6, 9 and 11 and added information for new INLINE™ 7 data link.
Date	13-Dec-2016
Section	Section F - Familiarization
Procedure Number	00-022-999 (/qs3/pubsys2/xml/en/procedures/00/00-022-999.html)
Procedure Title	Service Tools and Hardware - Overview
Update Reason	Removed or corrected information for INLINE™ 4, 5, and 6. Added new information for INLINE™ 7 data link.
Date	09-Aug-2018
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc294 (/qs3/pubsys2/xml/en/procedures/87/87-fc294.html)
Procedure Title	Auxiliary Temperature Sensor Input 1 Circuit - Voltage Below Normal or Shorted to Low Source
Update Reason	Added fault code to manual.

Date	09-Aug-2018
Section	Section 19 - Electronic Controls - Group 19
Procedure Number	87-019-250 (/qs3/pubsys2/xml/en/procedures/87/87-019-250.html)
Procedure Title	Connector, 50-Pin
Update Reason	Added procedure to manual.
Date	14-Aug-2018
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-t05-173 (/qs3/pubsys2/xml/en/procedures/87/87-t05-173.html)
Procedure Title	Fuel Rack Actuator Driver Mechanically Stuck Closed - Data Valid But Below Normal Operating Range - Moderately Severe Level
Update Reason	Revised Step 2A to include escalation directions.
Date	28-Jan-2020
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-fc293 (/qs3/pubsys2/xml/en/procedures/87/87-fc293.html)
Procedure Title	Auxiliary Temperature Sensor Input 1 - Voltage Above Normal, or Shorted to High Source
Update Reason	Added fault code overview to the manual.
Date	28-Jan-2020
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	87-t05-293 (/qs3/pubsys2/xml/en/procedures/87/87-t05-293.html)
Procedure Title	Auxiliary Temperature Sensor Input 1 - Voltage Above Normal, or Shorted to High Source
Update Reason	Added troubleshooting tree to manual.

Date	12-Jul-2022
Section	Section 19 - Electronic Controls - Group 19
Procedure Number	87-019-050 (/qs3/pubsys2/xml/en/procedures/87/87-019-050.html)
Procedure Title	Fuel Shutoff Valve
Update Reason	Corrected graphic for step that calls for EHAB fuel shutoff valve to be disconnected from the engine harness.

Date	16-Aug-2022
Section	Section 19 - Electronic Controls - Group 19
Procedure Number	87-019-050 (/qs3/pubsys2/xml/en/procedures/87/87-019-050.html)
Procedure Title	Fuel Shutoff Valve
Update Reason	Removed incorrect graphic 05400020 and replaced it with the correct one 19a00249.

Date	20-Sep-2024
Section	Section 19 - Electronic Controls - Group 19
Procedure Number	87-019-031 (/qs3/pubsys2/xml/en/procedures/87/87-019-031.html)
Procedure Title	Engine Control Module
Update Reason	No changes made to procedure, needed to update procedure title to align to proper assigned title.

Date	11-Jul-2025
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666214-t092-5-tr (/qs3/pubsys2/xml/en/manual/3666214/3666214-t092-5-tr.html)
Procedure Title	Fuel Leak - External
Update Reason	Added troubleshooting symptom tree to manual.

Date	26-Aug-2025
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666214-t016-tr (/qs3/pubsys2/xml/en/manual/3666214/3666214-t016-tr.html)
Procedure Title	Communication Error - Electronic Service Tool or Control Device
Update Reason	Added cause and correction that a technical service bulletin is available.

Last Modified: 20-Nov-2025
